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Parametric studies of hypersonic laminar flow control using a porous coating of regular microstructure

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To aid in the design of an ultrasonically absorptive coating (UAC) to be tested on a 7-degree half-angle sharp cone in the CUBRC LENS I shock tunnel, parametric studies of the coating laminar-flow-control performance are conducted for Mach=7 and Mach=10 free-stream conditions. The second-mode amplification factors, *N*, are calculated using the reduced-order computational package that includes the compressible Blasius mean flow and the local-parallel linear stability solver. These *N*-factors agree well with those predicted by the STABL solver that opens up an opportunity to conduct quick turn-around computations of the UAC performance. Stability calculations are carried out for the uncoated (solid) and coated (porous) wall. A porous coating of regular microstructure, which comprises equally spaced vertical cylindrical blind micro-holes of fixed radius, spacing and depth, is analyzed. The UAC parameters, at which the coating massively suppresses the second mode and can lead to significant (more than twice) increase of the laminar run, are predicted. Estimates of the UAC roughness effect indicate that the coating can be treated as aerodynamically smooth in the unit Reynolds number range required for transition experiments. It is shown that the UAC performance strongly increases with porosity. In this connection, it is suggested to investigate a rectangular or honeycomb patterns, which allow for coatings of substantially higher porosity compared with pores of circular cross-section.

Nomenclature

A	=	porous layer admittance
h	=	thickness of porous layer
M	=	Mach number
N	=	amplification factor
r	=	pore radius
Re_1	=	unit Reynolds number
s	=	spacing
U	=	mean flow velocity
(u, v, w)	=	amplitudes of velocity disturbance components
p	=	pressure disturbance
(x, y, z)	=	Cartesian coordinates
(α, β)	=	wavenumber components
ν	=	kinematic viscosity
ρ	=	density
θ	=	temperature disturbance

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ω = angular frequency

Subscripts

e = upper boundary-layer edge

tr = transition onset

∞ = free stream

I. Introduction

Laminar-turbulent transition during hypersonic flight represents a severe restriction to vehicle performance,[1] since early transition causes a significant increase (by a factor of 3-8)[2,3] in heat transfer that translates to higher cost and weight of thermal protection system (TPS). For hypersonic air-breathers, premature transition may be mission-critical because it reduces propulsion system efficiency, increases viscous drag (that can be more than 30% of the total drag), as well as degrades aerodynamic control surfaces and reaction control system performance. Because of these issues, strategies for achieving economically viable aerospace systems require laminar flow control (LFC) concepts that substantially delay transition.[4] State-of-the-art active and reactive LFC methods seem to be not practical under the severe conditions of hypersonic flight. Accordingly, passive LFC methods are promising to solve this problem.

In contrast to blunt configurations (e.g., Viking, Apollo type capsules) and moderately blunt lifting type configurations (e.g., Space Shuttle, the X-37 reusable launch vehicle), air-breathing hypersonic vehicles (e.g., NASP, Hyper-X) with relatively sharp leading edges can achieve high lift-to-drag ratio and fly at low angles of attack. This significantly increases local Mach numbers on the vehicle surfaces and leads to favorable changes in laminar-turbulent transition, namely:

- Roughness-induced transition becomes less important compared to other transition mechanisms since the critical roughness height quickly increases with the local Mach number[5]
- Relatively small leading-edge radii associated with hypersonic airbreathing cruise vehicles prevents transition due to nosetip roughness and leading-edge contamination[6]
- The first mode (associated with Tollmien-Schlichting waves) does not amplify because of low wall temperature ratios[7]

Furthermore, shaping of hypersonic air-breathers is designed to achieve massive reduction of cross-flow and Görtler vortices. Under such conditions the second mode may become a dominant instability. This suggests that the laminar run can be substantially increased by stabilization of the second mode.

Because the second mode represents high-frequency (ultrasonic) trapped acoustic waves, Malmuth *et al.*[8] assumed that a passive ultrasonically absorptive coating (UAC) may stabilize the second mode and, at the same time, be aerodynamically smooth. This hypothesis was confirmed by the theoretical study[9] based on the linear stability theory. The transition experiments of Rasheed *et al.*,[10] which were performed on a sharp cone in the GALCIT T5 high-enthalpy shock tunnel, qualitatively agree with the theoretical predictions. The stability experiments[11,12] were conducted on sharp cones with porous coatings of regular and random microstructures in the T-326 wind tunnel of the Institute of Theoretical and Applied Mechanics (ITAM, Novosibirsk). Hot wire measurements and linear stability theory (LST) showed that the UAC of fibrous absorbent material (felt metal) effectively stabilizes the second mode and slightly destabilizes the first mode.[11] The UAC of regular microstructure (a perforated thin sheet which is analogous to that used in the experiments[10]) also leads to massive reduction of the second-mode growth rate, while it weakly affects the first mode.[12] Nonlinear aspects of the second mode stabilization by UAC of regular microstructure were studied using bi-spectral analysis.[13] It was shown that the harmonic resonance, which is quite pronounced in the latter stages of the disturbance evolution on solid surfaces, is completely suppressed on the porous surface. Furthermore, the experiments,[14] which were carried out in the Mach 12 high-enthalpy wind tunnel, showed that the felt-metal coating significantly (up to 100% of the model length) delayed transition.

Two-dimensional direct numerical simulation (DNS) of the UAC performance was conducted for near-wall flows over a flat plate, sharp cone and compression corner at free-stream Mach numbers 5-6.[15] Numerical data obtained for disturbances propagating in the boundary layer on a flat plate agree satisfactory with LST. It was found that the coating end effects, which are due to discontinuity of the boundary conditions at the juncture between solid and porous walls, are localized over 2-3 disturbance wavelength and can be neglected in calculations of the integral UAC performance. The DNS demonstrates that the UAC weakly affects acoustic disturbances and the initial amplitudes of boundary-layer modes associated with receptivity problem, while it strongly suppresses the second-mode amplification. For the compression corner flow, the DNS showed that the porous coating weakly affects high-

frequency disturbances in the separation region and strongly stabilizes them in the reattached boundary layer downstream from the separation bubble.

These studies confirmed robustness of the UAC stabilization concept and motivated further experimental, DNS and theoretical efforts to mature the UAC-LFC methodology and provide a basis for design of UAC-TPS test articles that could be manufactured and tested within the framework of the HIFiRE (Hypersonic International Flight Research and Experimentation) program.[16]

The aforementioned stability and transition experiments have been carried out on relatively small models. It is important to demonstrate robustness of the UAC-LFC method on large configurations. In this connection, it is planned to carry out transition experiments in the CUBRC LENS facilities on a 3 m length cone model instrumented with the UAC surface. To aid in the design of the porous coating to be tested in these experiments, we have conduced parametric studies of the UAC-LFC performance for the free-stream conditions of the CUBRC LENS I shock tunnel. Results of this analysis are discussed hereafter.

II. Theoretical model

Herein we briefly discuss the theoretical model used for parametric estimates of the UAC-LFC performance. Consider the laminar boundary-layer flow past a sharp cone at zero angle of attack. The fluid is treated as a perfect gas of specific heat ratio $\gamma = 1.4$ and Prandtl number $\mathrm{Pr} = 0.72$. The mean flow in the laminar boundary layer is calculated using a self-similar solution of boundary-layer equations (compressible Blasius solution). It is assumed that the mean flows on both porous and solid surfaces are identical. The viscosity μ is calculated using the Sutherland formula

$$\mu^* = \mu_e^* T^{3/2} \frac{1+S}{T+S}, \; S = \frac{110.4}{T_e^*}, \; T \equiv \frac{T^*}{T_e^*}. \tag{1}$$

where T is temperature and asterisks denote dimensional quantities. For stability calculations, the second viscosity is approximated as $\mu_v = 0.8\mu$. The coordinates x, y, z are nondimensionalized using the boundary-layer length scale $\Delta^* \equiv \sqrt{\nu_e^* x_p^* / U_e^*}$, where $x_p^* = x^* / 3$ according to the Mangler transformation. Three-dimensional disturbances are represented in the traveling-wave form

$$\tilde{q} = \mathrm{Real}\{q(y)\exp[i(\alpha x + \beta z - \omega t)]\}, \tag{2}$$
$$\tilde{q} = (\tilde{u}, \tilde{v}, \tilde{w}, \tilde{p}, \tilde{\theta}),$$

where $\tilde{u}$, $\tilde{v}$, $\tilde{w}$ are velocity components referenced to U_e^*; $\tilde{p}$ is pressure referenced to $\rho_e^* U_e^{*2}$, $\tilde{\theta}$ is temperature referenced to T_e^*; $\alpha = \alpha^* \Delta^*$ and $\beta = \beta^* \Delta^*$ are wavenumber components; $\omega = \omega^* \Delta^* / U_e^*$ is angular frequency. The linear system of stability equations is written for a locally parallel boundary layer as

$$\frac{d\mathbf{z}}{dy} = \mathbf{S} \cdot \mathbf{z}, \tag{3}$$
$$\mathbf{z} = (u, du/dy, v, p, \theta, d\theta/dy, w, dw/dy)^T.$$

Here $\mathbf{S}$ is an 8×8 matrix with elements being functions of the mean flow profiles, Reynolds number, and the disturbance characteristics ω, α, β.[17] On the solid wall, the boundary conditions are

$$u(0) = v(0) = w(0) = \theta(0) = 0. \tag{4}$$

Since the boundary-layer modes exponentially decay as $y \to \infty$, the outer boundary conditions are

$$u(\infty) = v(\infty) = w(\infty) = \theta(\infty) = 0. \tag{5}$$

Solutions of the eigenvalue problems (3)-(5) give the dispersion relationship $F(\alpha, \beta, \omega) = 0$. For spatial instability in 2-D or conical boundary layers, the frequency ω and the transverse wavenumber β are real, whereas α is complex eigenvalue. If $\mathrm{Im}(\alpha) < 0$, then the disturbance amplifies with the spatial growth rate $\sigma_\alpha = -\mathrm{Im}(\alpha)$.

Consider a porous coating with equally spaced vertical cylindrical blind micro-holes (figure 1). This coating has a regular microstructure that is characterized by the pore radius r and spacing s. On the porous wall, the boundary conditions for the vertical velocity and temperature disturbances are expressed as[9]

$$v(0) = Ap(0)\,, \tag{6a}$$

$$\theta(0) = Bp(0)\,, \tag{6b}$$

where A (porous-layer admittance) and B (thermal admittance) are complex quantities that depend on the porous layer characteristics, mean-flow parameters at $y = 0$ as well as the disturbance frequency and wavelength. The

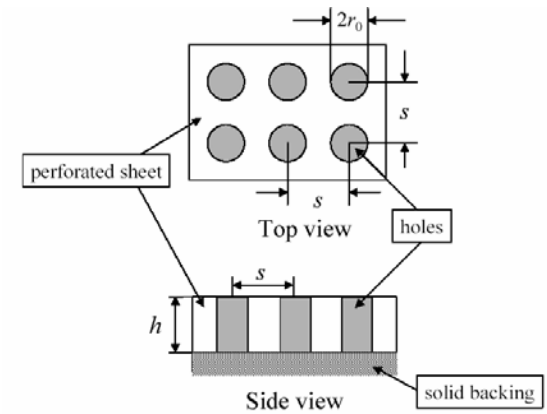


Figure 1. Porous coating with regular microstructure.

admittance A is calculated as[9]

$$A = -\frac{\phi}{Z_0}\tanh(\Lambda h)\,, \tag{7}$$

where $h = h^* / \Delta^*$ is nondimensional thickness of the porous layer, $\phi = \pi r^2 / s^2$ is porosity. The characteristic impedance Z_0 and the propagation constant Λ are expressed in terms of the nondimensional complex dynamic density $\tilde{\rho} = \tilde{\rho}^* / \rho_e^*$ and the complex compressibility $\tilde{C} = \tilde{C}^* p_e^*$ as

$$Z_0 = \frac{\sqrt{\tilde{\rho} / \tilde{C}}}{M_e \sqrt{T_w}}\,,\ \Lambda = \frac{i\omega M_e}{\sqrt{T_w}}\sqrt{\tilde{\rho}\,\tilde{C}}\,, \tag{8}$$

where $T_w = T_w^* / T_e^*$ is nondimensional wall temperature. The values of $\tilde{\rho}^*$ and $\tilde{C}^*$ are calculated using analytical solutions for acoustic disturbances propagating within a long tube as reported in Ref. 18. Note that these solutions account for gas rarefaction effect for small but finite Knudsen numbers based on the pore radius. As shown in Ref. 9 the thermal admittance produces negligible effect on the second-mode stability that allows us to put $B = 0$.

To formulate the wall boundary conditions for u and w, one should solve the problem near the pore opening, average the local solution over the UAC surface and establish relationships between $u(0)$, $w(0)$ and other disturbance components. This complicated procedure can be avoided using the following arguments. The second mode is inviscidly unstable and has maximum increments for 2-D waves ($\beta = 0$). In the limit $R \equiv \sqrt{U_e^* x^* / \nu_e^*} \to \infty$, the system of 2-D stability equations reads

$$\begin{aligned}
&-i\omega\rho + i\alpha\rho_s u + (\rho_s v)' = 0\,, \\
&\rho_s\left(i(\alpha U_s - \omega)u + U_s' v\right) + i\alpha p = 0\,, \\
&\rho_s i(\alpha U_s - \omega)v + p' = 0\,, \\
&\rho_s\left(i(\alpha U_s - \omega)\theta + T_s' v\right) = (\gamma - 1)M_e^2 i\left(\alpha U_s - \omega\right)p\,,
\end{aligned} \tag{9}$$

$$\frac{p}{P_s} + \frac{\theta}{T_s} + \frac{\rho}{\rho_s} = 0 \ ,$$

where the subscript "s" denotes the mean-flow quantities and prime stands for the derivative d/dy. This system is reduced to the well known system of equations for $v(y)$ and $p(y)$ [17]

$$v' = \frac{U_s'}{U_s - c} v + i\alpha \frac{T_s - M^2 (U_s - c)^2}{U_s - c} p \ , \tag{10}$$

$$p' = -i\alpha \frac{U_s - c}{T_s} v \ ,$$

where $c = \omega / \alpha$ is phase velocity. Using the second equation of (9) and (10) we obtain on the wall surface

$$v(0) = -i \frac{T_{sw}}{\omega} p'(0) \ , \tag{11}$$

$$u(0) = -\frac{T_{sw} U_{sw}'}{\omega^2} p'(0) - \frac{T_{sw}}{c} p(0) \ . \tag{12}$$

The boundary condition (11) can be written in the form (6a), while the longitudinal velocity component is expressed as

$$u(0) = -\left[i U_{sw}' v(0) + \alpha T_{sw} p(0) \right] / \omega \ . \tag{13}$$

The viscous effects are essential in the near-wall Stokes layer (where the longitudinal velocity is quickly adjusted to the boundary condition $u(0) = 0$) and in the critical layer located in the vicinity of critical point $y_c : U_s(y_c) = c_r$, where subscript s denotes real quantity. To illustrate this feature we consider the compressible Blasius profiles of the mean flow on a flat plate at $M_e = 6.21$ and $T_{sw} = 1$ shown in figure 2, where $y = y^* / \sqrt{\nu_e^* x^* / U_e^*}$. The temperature eigenfunction $|\theta(y)|$ of the 2-D second-mode wave of frequency $F \equiv \omega^* \nu_e^* / U_e^{*2} = 6 \times 10^{-5}$ is shown in figure 3 at the station corresponding to $R = 3452.54$. The vertical line shows the Stokes layer edge that is calculated as

$$\bar{\delta}_s \equiv \delta_s / \sqrt{\nu_e^* x^* / U_e^*} = \sqrt{\nu_w^* / \omega^*} / \sqrt{\nu_e^* x^* / U_e^*} = F^{-1/2} R^{-1} \sqrt{\nu_w^* / \nu_e^*} \ . \tag{14}$$

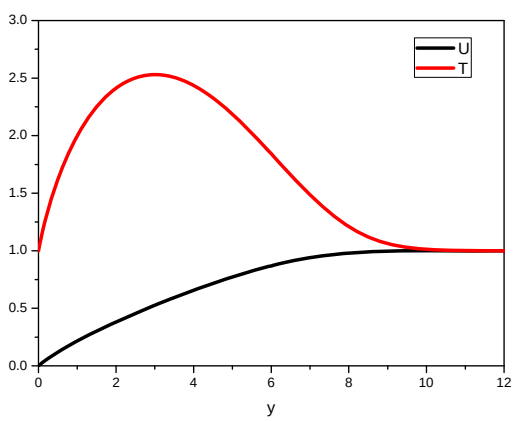


Figure 2. Mean flow profiles $U_s(y)$ and $T_s(y)$ calculated using the compressible Blasius solution, $M_e = 6.21$ and $T_{sw} = 1$.

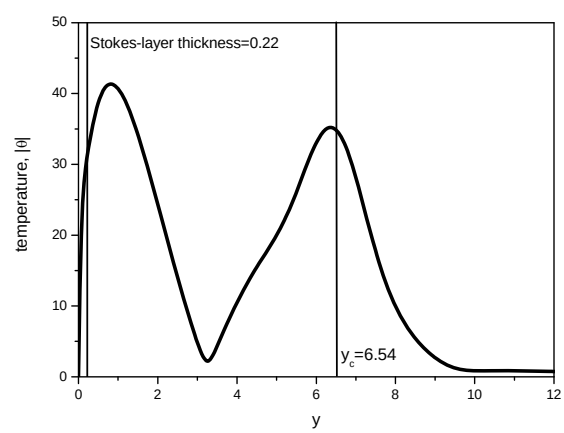


Figure 3. Eigenfunction $|\theta(y)|$ of 2-D second-mode wave, $F = 6 \times 10^{-5}$, $\alpha = (2.27157\text{E-}1, -8.17745\text{E-}3)$, solid wall.

In the case of $T_w^* = T_e^*$ considered herein, we have $\sqrt{\nu_w^* / \nu_e^*} = 1$, and the nondimensional thickness of Stokes layer is $\overline{\delta}_s \approx 0.22$. The ratio of $\overline{\delta}_s$ to the wavelength $\lambda = 2\pi / \alpha_r$ is approximately 0.008; i.e., the Stokes layer is very thin. The phase speed $c_r \approx 0.91$ gives the critical point $y_c \approx 6.54$ that correlates with the disturbance temperature maximum. The critical layer thickness is of the order of half-width of this maximum, $\overline{\delta}_c \approx 2$, that is 10 times larger than the Stokes-layer thickness.

These findings suggest that the viscous effect on the second mode instability is predominately due to the critical layer while the Stokes layer plays a minor role. To verify this assumption we conduct calculations of the second-mode wave by solving the stability equations (including all viscous terms in order to capture viscous effects in the critical layer) with the inviscid boundary condition (13) for $u(0)$ and $v(0) = \theta(0) = 0$. Figures 4 show comparisons of the eigenfunctions calculated with viscous and inviscid boundary conditions at the aforementioned parameters. As expected, these eigenfunctions agree very well. The eigenvalues are also very close:

$$\alpha_{\text{solid,vis}} = (2.272\text{E-1, -8.177E-3}),\ \alpha_{\text{solid,invis}} = (2.271\text{E-1, -8.200E-3}). \tag{15}$$

Discrepancy between $|u(y)|$ is appreciable in the Stokes layer only as shown in the left top panel of figure 4.

Similar calculations were conducted for the porous wall of $r^* = 25$ μm, $s^* = 100$ μm and $h^* = 300$ μm using:

- Inviscid boundary condition (13), $v(0) = Ap(0)$, $\theta(0) = 0$ (16)
- Simplified viscous boundary condition $u(0) = 0$, $v(0) = Ap(0)$, $\theta(0) = 0$ (17)

The boundary condition (13) corresponds to the case of maximum slip on the porous surface, while the boundary condition $u(0) = 0$ – to the case of no slip. In reality, the disturbance amplitude $u(0)$ lies somewhere in between. Comparing these extreme cases we can estimate sensitivity of the UAC performance to uncertainty of the boundary condition for u. As in the solid wall case, the eigenvalues for the plate with UAC are close to each other:

$$\alpha_{\text{porous,vis}} = (2.283\text{E-1, -2.689E-3}),\ \alpha_{\text{porous,invis}} = (2.283\text{E-1, -2.671E-3}).$$

The corresponding eigenfunctions $|u(y)|$ are shown in figure 5. The discrepancy between $|u(y)|$ is appreciable in the Stokes layer only.

The transition onset point, x_{tr}, is estimated using the e^N-method.[19] For 2-D waves, the amplification factors $N(x,\omega)$ are calculated at fixed dimensional frequencies ω^* as

$$N(x,\omega^*) = \int_{x_0(\omega^*)}^{x} \sigma_\alpha(x,\omega^*)dx, \tag{18}$$

where $x_0(\omega^*)$ is point at the lower neutral branch. The *N*-factor is correlated with the available transition onset data x_{tr} as $N(x_{tr},\omega^*) = N_{tr}$. The critical value of N_{tr} has been shown to be about 8-11 for many different smooth-body geometries in low-disturbance free-stream conditions.[19-21] The transition onset data in noisier environments, such as conventional wind tunnels and shock tunnels, are correlated with $N_{tr} \approx 5.5$.[22-24]

The foregoing analysis leads to the assumption that *N*-factors on porous and solid walls can be calculated with sufficient accuracy using the inviscid boundary conditions (16) or the simplified viscous boundary conditions (17). To validate this assumption, calculations of *N*-factors were carried out for a 7° half-angle sharp cone at zero angle of attack at the free-stream parameters corresponding to the CUBRC LENS I shock tunnel: $\mathrm{M}_\infty = 7.14$, $T_\infty^* = 231.8$ K. Calculations of the mean flow using the Navier-Stokes solver gives the local flow parameters $M_e = 6.21$, $T_{sw} = 1$. Distributions of *N*-factors on the solid surface and on the porous surface with UAC of $r^* = 25$ μm, $s^* = 100$ μm and $h^* = 300$ μm, are shown in figure 6. The solid lines correspond to the solution obtained with the inviscid boundary conditions (16), symbols – with the boundary conditions (17). As expected, the difference between $N(x)$ is negligible. These findings allow us to conduct parametric studies of the UAC laminar-flow-control performance using the boundary conditions (17) on the porous wall.

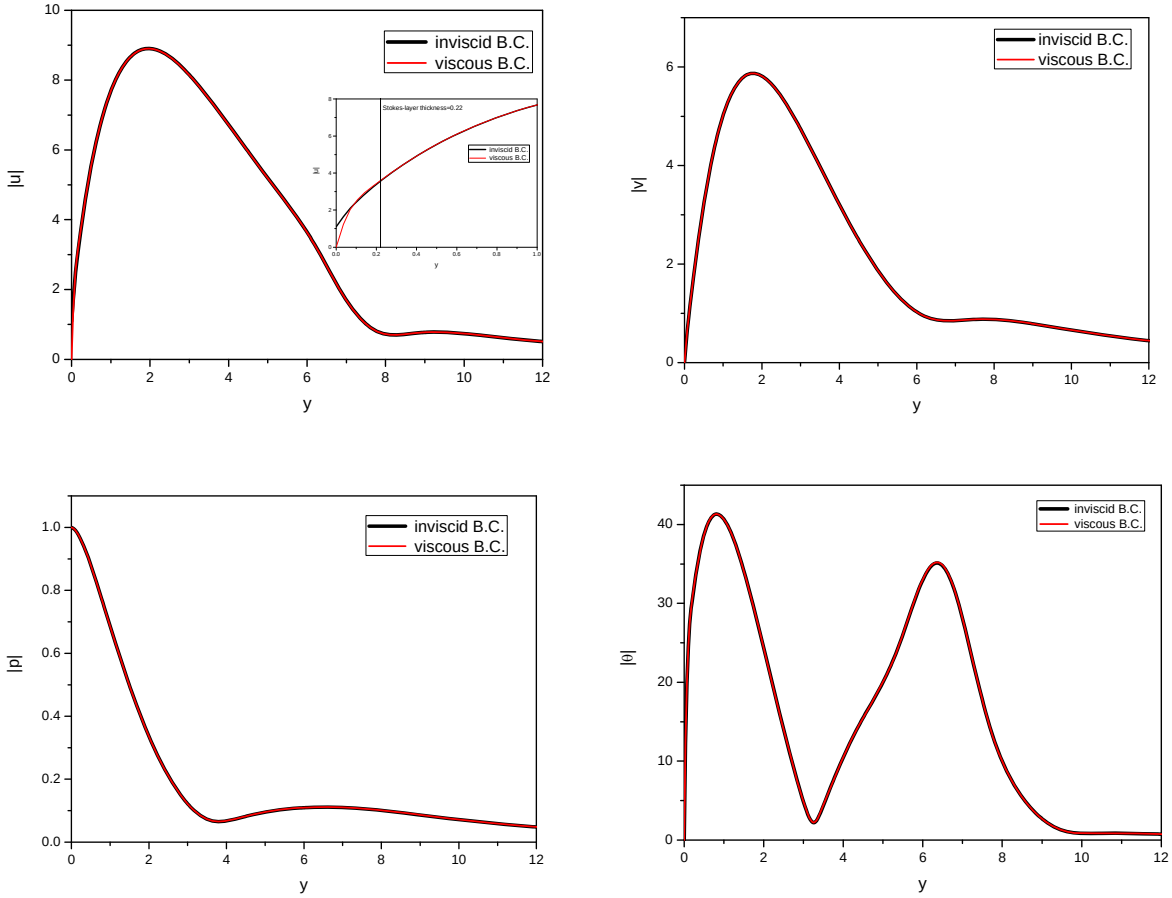


Figure 4. Eigenfunctions of 2-D second-mode wave of frequency $F = 6 \times 10^{-5}$; black lines – inviscid boundary conditions, red lines – viscous boundary conditions; solid wall.

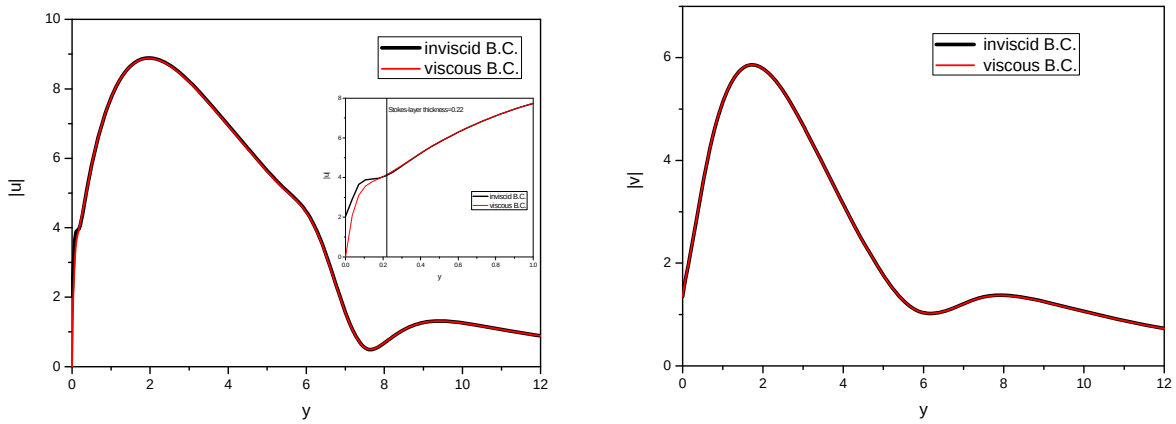


Figure 5. Eigenfunctions of 2-D second-mode wave of frequency $F = 6 \times 10^{-5}$; black lines – inviscid boundary conditions (16), red lines – simplified viscous boundary conditions (17); porous wall.

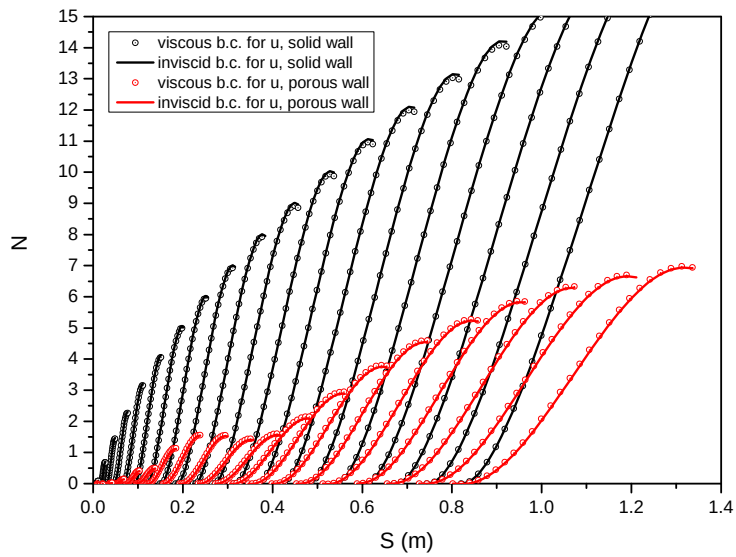


Figure 6. *N*-factors of 2-D second-mode; lines – inviscid boundary conditions (16), symbols – simplified viscous boundary conditions (17), S – longitudinal coordinate measured along the cone surface.

III. Transition on solid wall and comparison with STABL

To calibrate our stability and transition prediction tools, we have analyzed transition data of Ref. 22. Namely, we consider the laminar boundary-layer flow past a sharp 7-degree half-angle cone at zero angle of attack. The free-stream parameters correspond to Run 22 reported in Table 2 of Ref. 22: Mach number $\mathrm{M}_\infty = 10.04$, unit Reynolds number $\mathrm{Re}_{1\infty} = 1.92\times10^6$ m^{-1}, temperature $T^*_\infty = 198.62$ K. The cone length measured along the cone surface is $L^* = 2.2$ m, and the wall temperature is $T^*_w = 297.28$ K. The local flow parameters (at the upper boundary-layer edge) are determined from the Navier-Stokes solution for laminar flow: $\mathrm{Re}_{1e} = 3.2\times10^6$ m^{-1}, $T_w / T_e = 1.03$, $T^*_e = 297.93$ K and $M_e = 8.1$. At these parameters the compressible Blasius solution agrees well with that predicted by the Navier-Stokes solver (figures 7 and 8); i.e., the viscous-inviscid interaction weakly affects the boundary-layer mean flow.

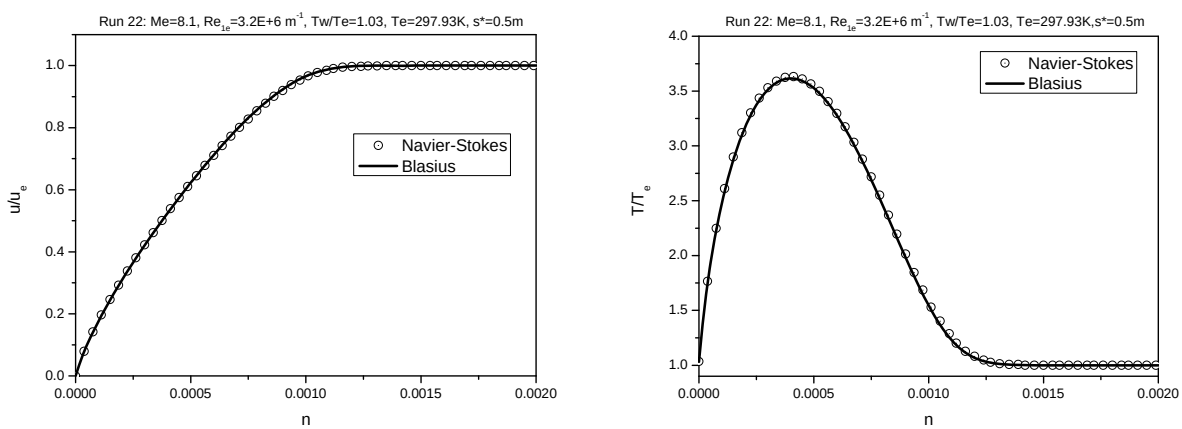


Figure 7. Flow velocity (left) and temperature (right) profiles at the station $s^* = 0.5$ m, here s^* – longitudinal coordinate measured along the cone surface, $n = n^* / L^*$ – coordinate normal to the cone surface.

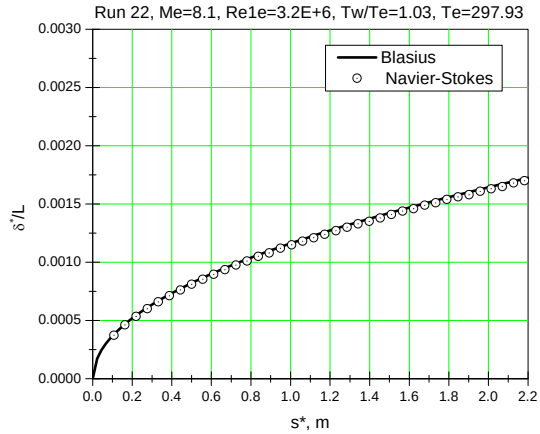


Figure 8. Boundary-layer displacement thickness versus longitudinal coordinate, s^*, measured along the cone surface.

Figure 9 shows the data scanned from Fig. 7 of Ref. 22. Symbols correspond to the heat flux measurements on the sharp cone model tested in the CUBRC LENS I shock tunnel (Run 22). The thin black line represents the laminar heat flux distribution predicted by the Navier-Stokes solver of STABL. The green dashed line shows an envelope of *N*-factors predicted by the PSE solver of STABL. The family of solid black lines shows the amplification factors predicted by our reduced-order computational package that includes the compressible Blasius mean flow and the local-parallel linear stability solver. Strikingly, an envelope of these amplification curves is very close to that predicted by STABL. This opens up an opportunity to conduct quick turn-around parametric computations of the transition onset points and search for the UAC parameters appropriate for further experimental and DNS studies.

The critical amplification $N_{tr} = 5.5$ was suggested in Ref. 22 for the CUBRC LENS I shock tunnel (see dashed black line in figure 9). This value is used for assessments of the transition locus on porous and solid surfaces.

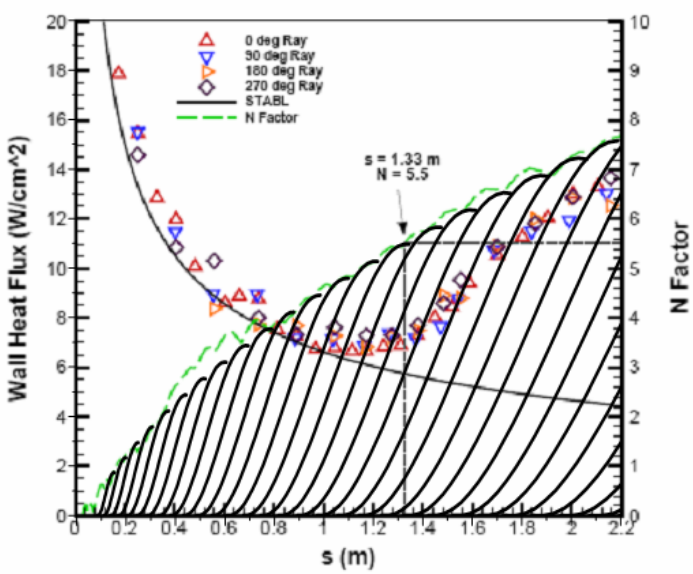


Figure 4. Comparison of *N*-factors predicted by STABL (dashed green line) with our calculations (family of solid black lines) for Run 22 (sharp cone); symbols – experimental measurements of the wall heat flux, thin black line – laminar heat flux predicted by STABL; s – longitudinal coordinate measured along the cone surface.

IV. Assessments of UAC performance

A. Free stream at Mach 7

Parametric calculations of the UAC-LFC performance are carried out for the free-stream conditions of Run 5 (see Table 1 of Ref. 22): $\mathrm{M}_\infty = 7.14$, $\mathrm{Re}_{1\infty} = 9.76 \times 10^6$ m^{-1}, $T_\infty^* = 231.8$ K, the wall temperature is

$T_w^* = 299.3$ K. The cone length (measured along the cone surface) is $L^* = 2.2$ m. The Navier-Stokes solution gives the local parameters: $\mathrm{Re}_{1e} = 14.2 \times 10^6$ m^{-1}, $T_e^* = 299.02$ K, $M_e = 6.21$, and the wall temperature ratio $T_w / T_e = 1$. It is assumed that M_e, T_w / T_e and T_w^* do not change in the range of unit Reynolds number considered hereafter. For a certain value of Re_{1e}, stability calculations are conducted for the second-mode 2-D waves.

Preliminary estimates indicated that the UAC of $r^* = 25$ μm, $s^* = 100$ μm and $h^* = 10r^* = 250$ μm (porosity $\phi = 0.196$) are appropriate for the aforementioned flow conditions. Figures 10 and 11 show the amplification factors $N(X^*, \omega_j^*)$ for various disturbance frequencies ω_j^* as well as the ratio of the disturbance wavelength to the pore spacing λ^* / s^*. The latter characterizes fineness of the UAC structure. Hereafter, X^* is the longitudinal coordinate measured from the cone tip along the cone surface. The horizontal line corresponds to the critical value $N_{tr} = 5.5$.

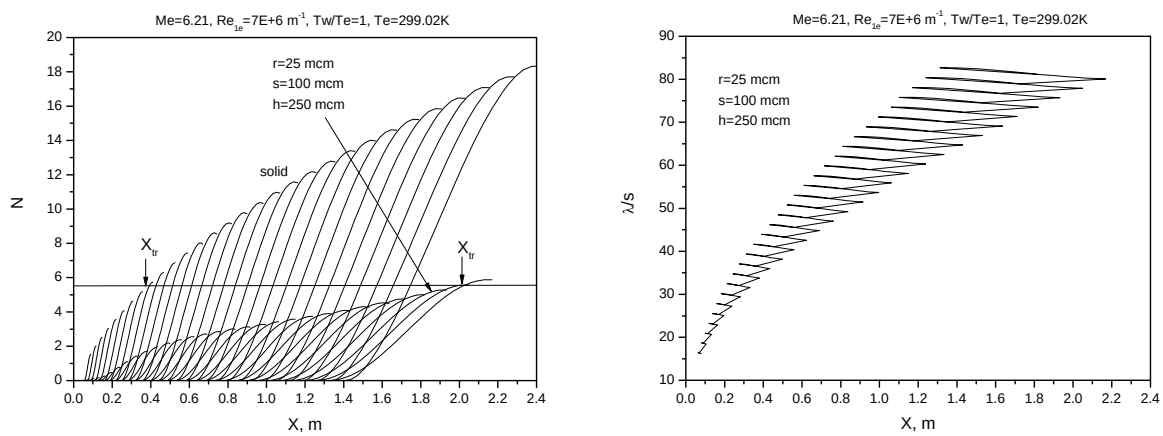


Figure 10. *N*-factors for solid and porous walls (left) and the coating fineness distributions λ^* / s^* (right) along the cone surface, $\mathrm{Re}_{1e} = 7 \times 10^6$ m^{-1}, $\mathrm{M}_\infty = 7.14$.

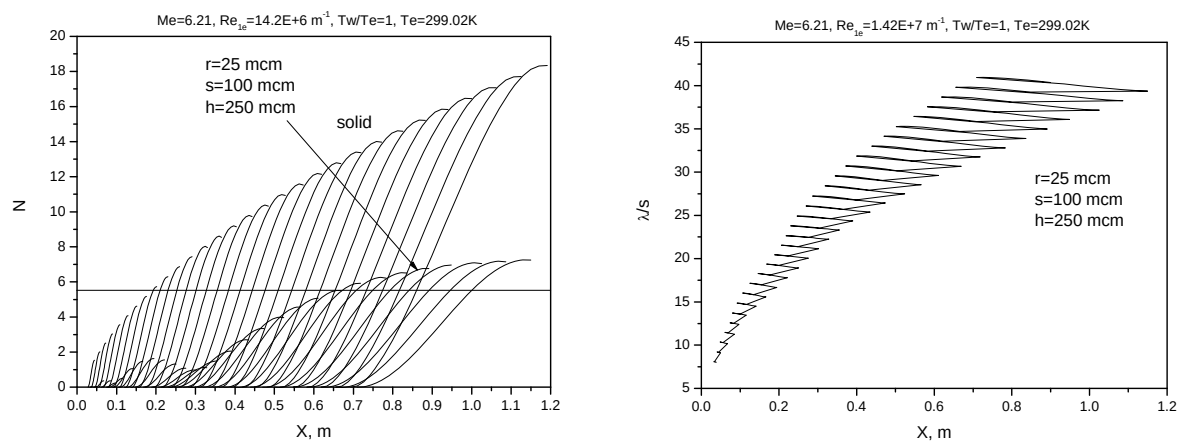


Figure 11. *N*-factors for solid and porous walls (left) and the coating fineness distributions λ^* / s^* (right) along the cone surface, $\mathrm{Re}_{1e} = 14.2 \times 10^6$ m^{-1}, $\mathrm{M}_\infty = 7.14$.

Similar calculations were conducted in the range of local unit Reynolds numbers $2\times10^6 \le \mathrm{Re}_{1e} \le 20\times10^6$ m^{-1}. Results are summarized in Figure 12, where the transition onset point is shown as a function of Re_{1e} for the solid (black line) and porous (red line) wall. The theory predicts that the UAC strongly stabilizes the second mode and can lead to the laminar run increase on the order of magnitude (see data close to $\mathrm{Re}_{1e} \approx 10^7$ m^{-1}). The UAC performance is not monotonic versus Re_{1e}. Namely, the transition onset point on the porous wall has a jump at $\mathrm{Re}_{1e} \approx 11.7\times10^6$ m^{-1}. This behavior is associated with a finite depth of pores. The disturbances penetrating into a blind hole are reflected by the hole bottom that affects the UAC admittance. As a result, the envelope of amplification curves has modulations (see the left plot in figure 11). As the unit Reynolds number increases crossing the value $\mathrm{Re}_{1e} \approx 11.7\times10^6$ m^{-1}, the envelope first maximum attains the critical level $N = 5.5$, and the predicted transition onset point jumps toward the cone tip. This trend is illustrated in figure 13, where the amplification curves are shown in the vicinity of this jump.

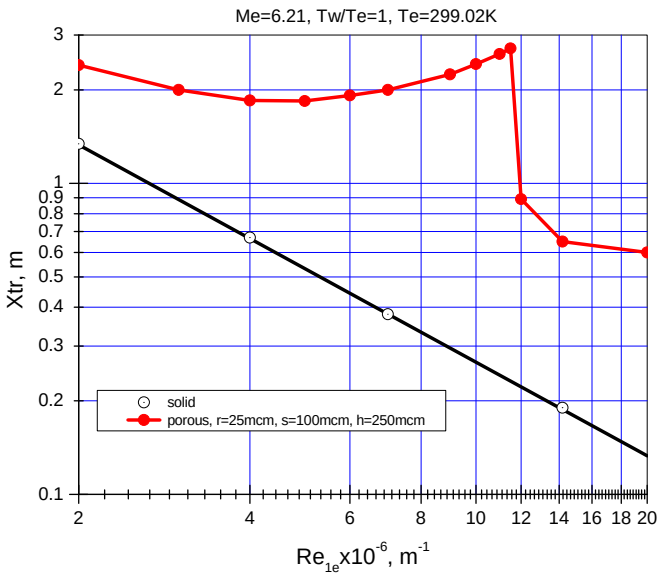


Figure 7. The transition onset point X_{tr} as a function of Re_{1e} for solid (black line with symbols) and porous (red line with symbols) walls, $\mathrm{M}_\infty = 7.14$.

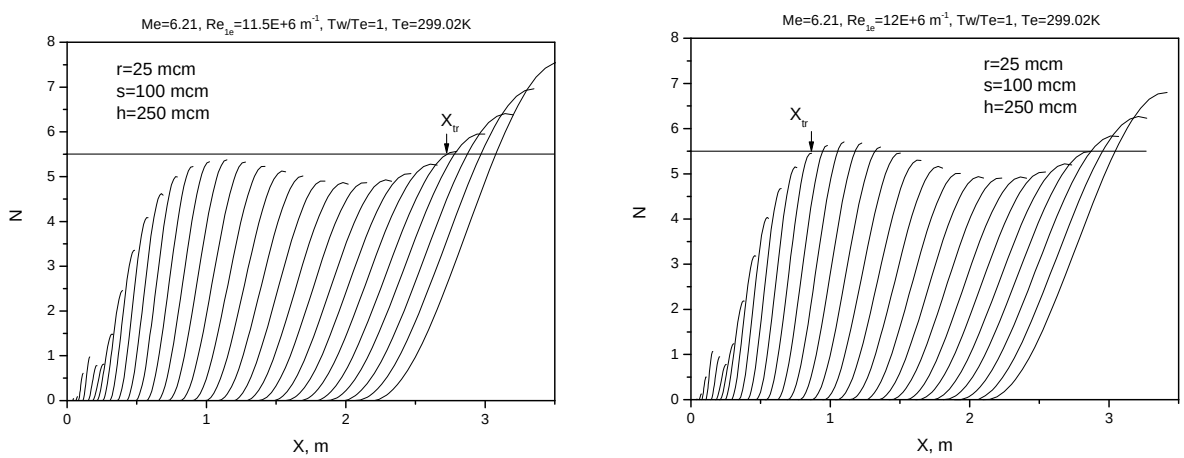


Figure 13. *N*-factors for porous wall at $\mathrm{Re}_{1e} = 11.5\times10^6$ m^{-1} (left) and $\mathrm{Re}_{1e} = 12\times10^6$ m^{-1} (right).

The foregoing stability analysis lies in the expectation that the porous surface is aerodynamically smooth in order for the proposed mechanism to effectively delay transition. If this were not the case, then the holes would act as distributed surface roughness and prematurely trip the boundary layer. Unfortunately, data relevant to roughness-induced transition on surfaces with distributed blind holes are very limited. There is only one set of experimental

data obtained by Rasheed et al.[10] in the GALCIT T5 shock tunnel on a 5-degree half-angle sharp cone at zero angle of attack. In these experiments the cone model had a smooth surface over its half and the aforementioned coating over the other half beginning at 148 mm from the cone tip. The UAC had 60 μm diameter holes spaced 100 μm apart in a rectangular grid arrangement resulting in a porosity of 28%. In Ref. 10, the coating roughness is characterized by the Reynolds number

$$\mathrm{Re}_D^* = \frac{\rho^*(T_{ref}^*)U_e^* D^*}{\mu^*(T_{ref}^*)}, \qquad (19)$$

where $D^* = 2r^*$ is hole diameter, the density ρ^* and viscosity μ^* are calculated at the reference temperature, T_{ref}^*, determined as

$$\frac{T_{ref}^*}{T_e^*} = 0.5 + 0.039M_e^2 + 0.5\frac{T_w^*}{T_e^*}. \qquad (20)$$

Assuming that the viscosity coefficient is approximated as

$$\mu^*(T_{ref}^*) = \mu^*(T_e^*)(T_{ref}^* / T_e^*)^{0.75},$$

and using the relation

$$\rho^*(T_{ref}^*) = \rho^*(T_e^*)T_e^* / T_{ref}^*,$$

we get

$$\mathrm{Re}_D^* = \mathrm{Re}_{1e}\frac{D^*}{(T_{ref}^* / T_e^*)^{1.75}}. \qquad (21)$$

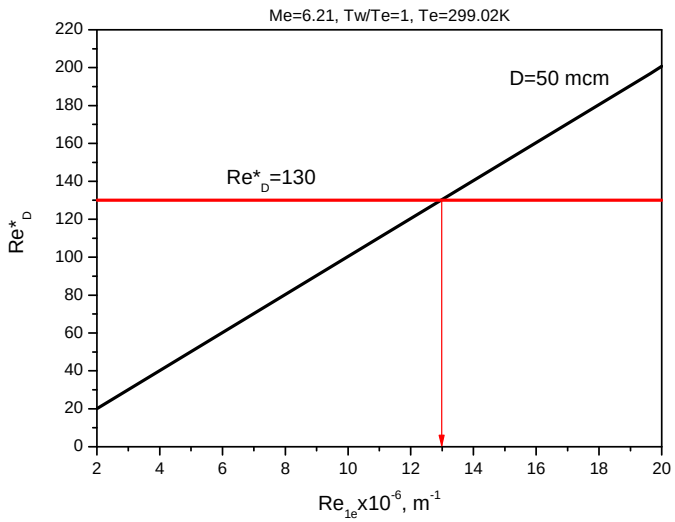


Figure 14. The Reynolds number Re_D^* versus Re_{1e}. The criterion $\mathrm{Re}_D^* = 130$ indicates that UAC can be treated as aerodynamically smooth for $\mathrm{Re}_{1e} < 13\times10^6$ m^{-1} and $\mathrm{M}_\infty = 7.14$.

The experiment[10] showed that the coating becomes less effective at $\mathrm{Re}_D^* \approx 130$ and prematurely trips the boundary layer at $\mathrm{Re}_D^* > 300$. Since the coating characteristics[10] are close to those considered herein, we assume that the criterion $\mathrm{Re}_D^* = 130$ can be used for estimates of the UAC roughness effect if the coating begins at a distance $X_0^* > 148$ mm from the cone tip. Figure 14 shows Re_D^* as a function of Re_{1e} for the UAC considered herein. The criterion $\mathrm{Re}_D^* = 130$ indicates that the UAC can be treated as aerodynamically smooth for $\mathrm{Re}_{1e} < 13\times10^6$ m^{-1}.

B. Free stream at Mach 10

The same porous coating ($r^* = 25$ μm, $s^* = 100$ μm, $h^* = 10r^* = 250$ μm, $\phi = 0.196$) was analyzed for the free-stream conditions: $M_\infty = 10.04$, $T_\infty^* = 198.62$ K, and $T_w^* = 297.28$ K relevant to Run 22 of Ref. 22. The local flow parameters, which were predicted by the Navier-Stokes solver at $\mathrm{Re}_{1\infty} = 1.92\times10^6$ m^{-1}, are: $M_e = 8.1$, $T_e^* = 297.93$ K, $T_w / T_e = 1.03$, $\mathrm{Re}_{1e} = 3.2\times10^6$ m^{-1}. It is assumed that M_e, T_w / T_e and T_w^* do not change in the range of unit Reynolds number considered hereafter. Figure 15 shows the amplification factors $N(X^*, \omega_j^*)$ and the ratio λ^* / s^* for various disturbance frequencies ω_j^* at $\mathrm{Re}_{\infty 1} = 12\times10^6$ m^{-1}. Similar calculations were carried out at various Re_{1e} and summarized in figure 16. Comparing this figure with figure 12, we conclude that the UAC is less effective than in the Mach=7 case. However, the laminar run increase is still significant. The transition onset point is more than doubled in the range of $8\times10^6 < \mathrm{Re}_{1e} < 20\times10^6$ m^{-1}. For $\mathrm{Re}_{1e} < 8\times10^6$ m^{-1}, the relative pore radius r^* / λ^* becomes small that leads to decreasing of the UAC LFC performance.

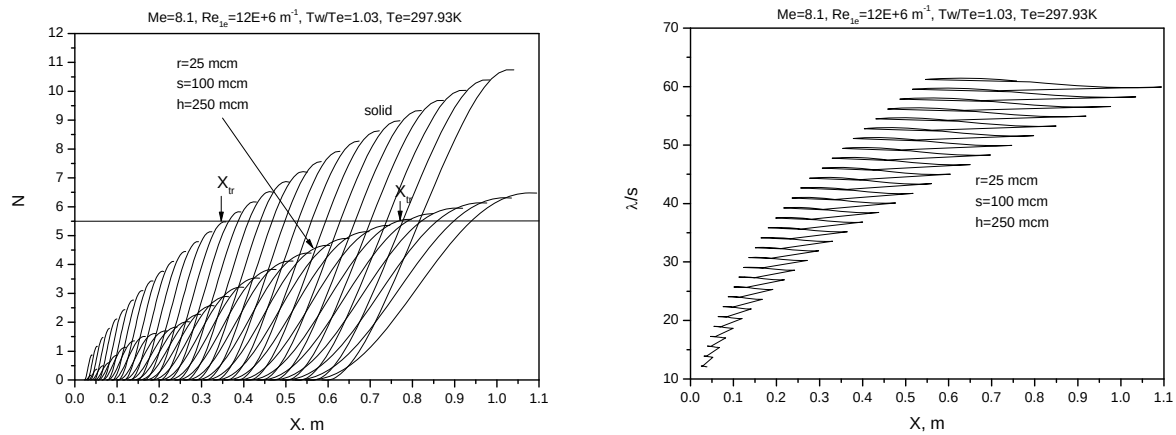


Figure 15. *N*-factors for solid and porous walls (left) and the coating fineness distributions λ^* / s^* (right) along the cone surface, $\mathrm{Re}_{1e} = 12 \times 10^6$ m^{-1}, $M_\infty = 10.04$.

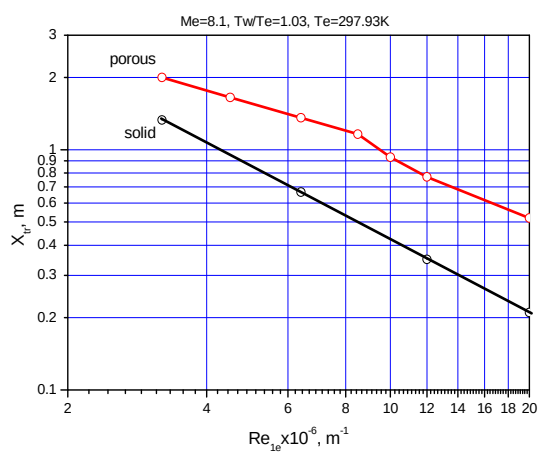


Figure 16. The transition onset point X_{tr} as a function of Re_{1e} for solid (black line with symbols) and porous (red line with symbols) walls, $M_\infty = 10.04$.

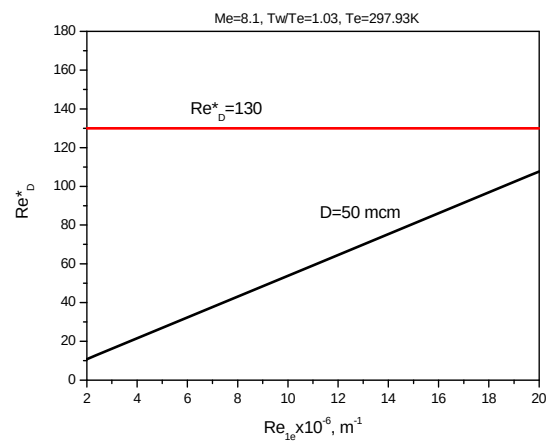


Figure 17. The Reynolds number Re_D^* versus Re_{1e}. The criterion $\mathrm{Re}_D^* = 130$ indicates that the UAC can be treated as aerodynamically smooth in the considered range of Re_{1e}, $M_\infty = 10.04$.

The Reynolds number Re_D^* is shown as a function of Re_{1e} in figure 17. It is seen that Re_D^* does not reach the critical value in the range of $2 \times 10^6 < \mathrm{Re}_{1e} < 20 \times 10^6$ m^{-1}; i.e., the coating is aerodynamically smooth in the case of $M_\infty = 10.04$.

C. UAC configuration for experiments

The foregoing calculations indicate that the porous coating of $r^* = 25$ μm, $s^* = 100$ μm and $h^* \approx 10r^*$ is appropriate for testing on a 7-degree half-angle sharp cone in the CUBRC LENS I shock tunnel at Mach=7 and Mach=10 free streams. Because of manufacturing constraints, the UAC thickness is specified as $h^* = 300$ μm ($h^* = 12r^*$). The UAC upstream boundary is placed at the station $X_0^* = 200$ mm. This configuration has been chosen for further consideration.

The transition onset points were calculated at $M_\infty = 7.14$ (figure 18). For the coating of $h^* = 300$ µm, the effect of disturbance reflection from the pore bottoms is weaker than in the case of $h^* = 250$ µm, and the transition onset point varies smoother versus Re_{1e} (compare blue and red curves). At $\mathrm{Re}_{1e} \approx 14 \times 10^6\,\mathrm{m}^{-1}$, transition occurs near the coating beginning, $X_0^* = 0.2$ m, the transition onset point jumps toward the cone tip and corresponds to the solid wall case. Figure 19 shows the Knudsen number $\mathrm{Kn} = l^* / r^*$ (where l^* is molecular mean free path) as a function of Re_{1e}. For the hole-radius $r^* = 25$ µm considered herein, the gas rarefaction is appreciable for $\mathrm{Re}_{1e} < 3.5 \times 10^6$ m^{-1}, where $\mathrm{Kn} > 0.2$. Similar calculations were conducted for the Mach number $M_\infty = 10.04$ (see figures 20 and 21).

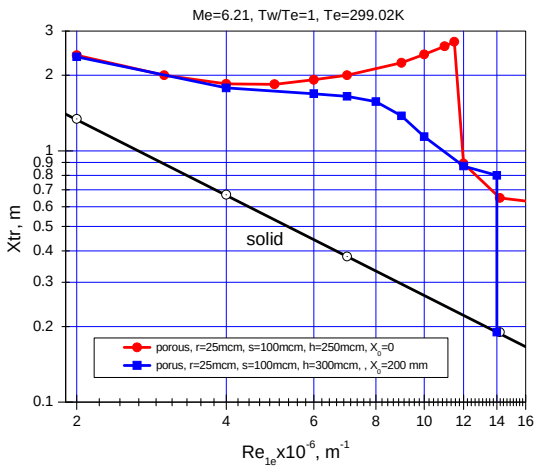


Figure 18. Transition onset point as a function of Re_{1e} for solid and porous walls, $M_\infty = 7.14$.

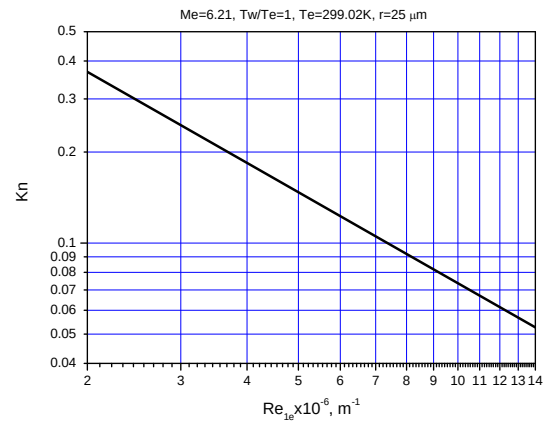


Figure 19. Knudsen number versus Re_{1e} at $M_\infty = 7.14$.

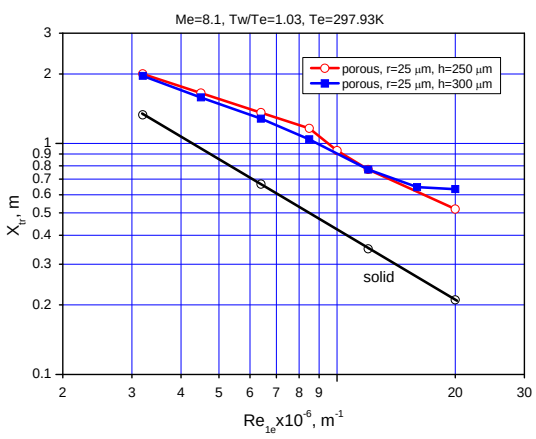


Figure 20. Transition onset point as a function of Re_{1e} for solid and porous walls, $M_\infty = 10.04$.

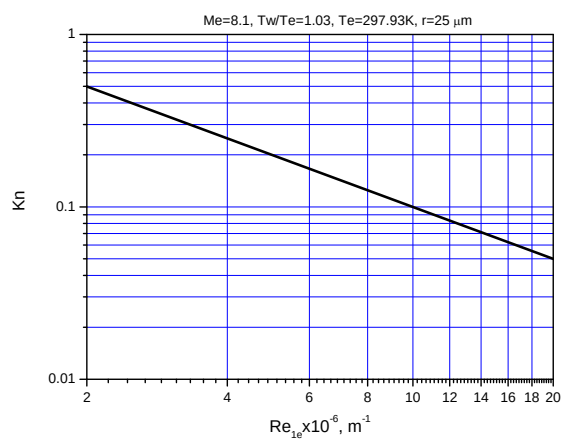


Figure 21. Knudsen number versus Re_{1e} at $M_\infty = 10.04$.

Transition assessments (figures 18 and 19) indicate that for experiments at Mach=7 the cone length should be at least 3 m to capture transition onset on the coated surface in the unit Reynolds number range of $2 \times 10^6 < \mathrm{Re}_{1e} < 12 \times 10^6$ m^{-1} (this range corresponds to $4.5 \times 10^6 < \mathrm{Re}_{1\infty} < 27 \times 10^6$ ft^{-1}). For experiments at

Mach=10, the cone length should be larger than 2 m to capture transition onset on the coated surface in the unit Reynolds number range of $4\times10^6 < \mathrm{Re}_{1e} < 20\times10^6\,\mathrm{m}^{-1}$ ($7.9\times10^6 < \mathrm{Re}_{1\infty} < 3.9\times10^7\,\mathrm{ft}^{-1}$). In accord with the data shown in figure 22, these ranges of $\mathrm{Re}_{1\infty}$ are achievable in the CUBRC LENS I shock tunnel.

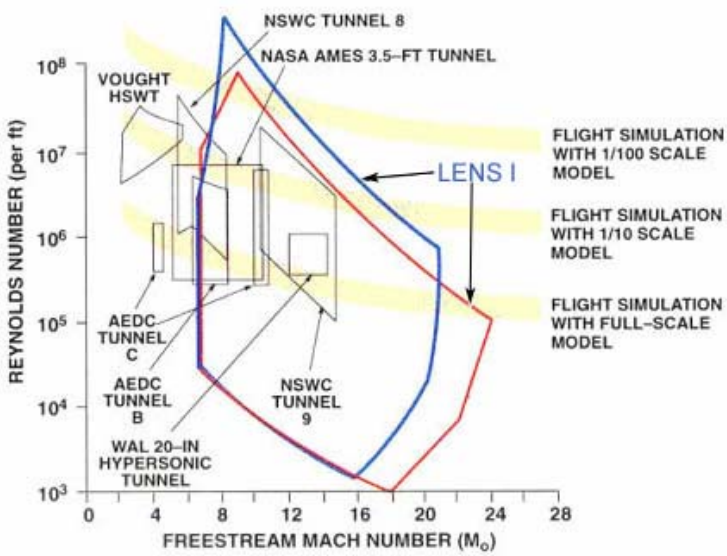


Figure 17. Free-stream parameters of the CUBRC LENS I shock tunnel.[22]

D. Porosity effect

Apparently the UAC-LFC performance increases with porosity. Figure 23 illustrates this trend for the coating of $r^* = 25$ μm and $h^* = 300$ μm. With decreasing the pore spacing from $s^* = 100$ μm ($\phi = 0.196$) to $s^* = 75$ μm ($\phi = 0.35$), the transition onset point increases from 1.25 m to 1.97 m.

For the UAC configuration shown in figure 1, the theoretical maximum $\phi = \pi/4 = 0.785$ corresponds to the minimal spacing $s^* = 2r^*$. More promising are configurations schematically shown in figure 24. For the square-rectangular or honeycomb patterns, it is feasible to fabricate coatings with porosity 70-90%.

In the theoretical model applied herein, the interaction between holes is neglected. Apparently, this interaction needs to be modeled especially for UAC of high porosity which comprises closely spaced holes depicted in figure 24. The structural constraints to these coatings should be formulated and verified experimentally.

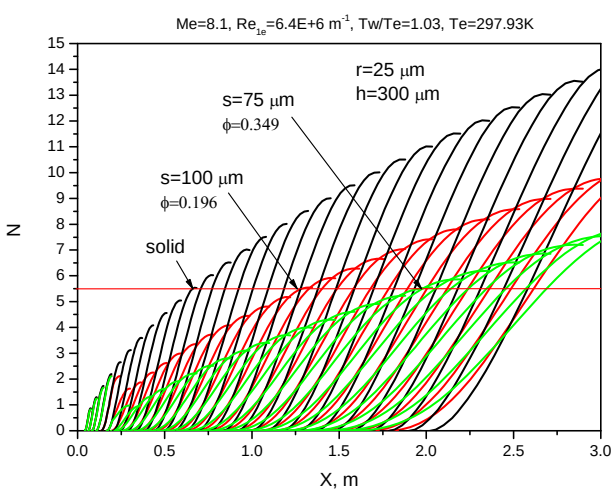
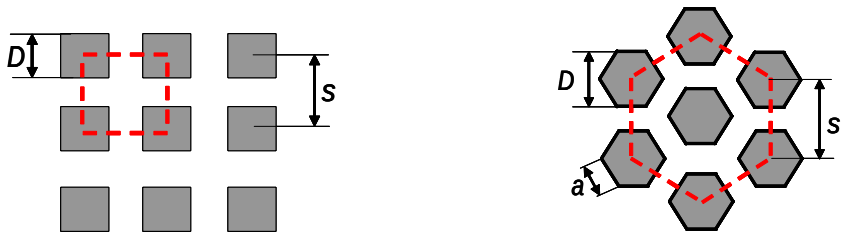


Figure 23. Porosity effect on the UAC-LFC performance, $\mathrm{Re}_{1e} = 6.4\times10^6$ **m^{-1},** $M_\infty = 10.04$**.**



Square/rectangular pattern

$$\phi = \frac{D^2}{s^2}$$

Honeycomb pattern

$$\phi \equiv 3\frac{a^2}{s^2} = \frac{D^2}{s^2}$$

Figure 24. Porosity of various UAC patterns, pore cross-sections are grey.

V. Summary

To aid in the design of the porous coating to be tested on a 7-degree half-angle sharp cone in the CUBRC LENS I shock tunnel, parametric studies of the UAC-LFC performance have been conduced for the Mach=7 and Mach=10 free-stream conditions.

The laminar flow calculations were performed for the cone at zero angle of attack using the Navier-Stokes solver. The mean-flow numerical solutions agree well with the compressible Blasius solution indicating that the viscous-inviscid interaction is negligible in the cases considered herein.

It was shown that the Stokes layer induced by the second mode near the wall is very thin and weakly affects the second-mode eigenvalues and eigenfunctions. This allows us to simplify the boundary conditions on porous wall for parametric studies of the UAC LFC performance.

To calibrate and verify our stability and transition prediction tools we analyzed the transition onset data[22] measured in the LENS I tunnel. Namely, we considered the laminar boundary-layer flow past a sharp 7-degree half-angle cone at zero angle of attack at Mach=10 free-stream conditions relevant to Run 22. Using our reduced-order computational package that includes the compressible Blasius mean flow and the local-parallel linear stability solver, we calculated *N*-factors for the second mode and compared them with the predictions of STABL reported in Ref. 22. Strikingly, our results agree very well with those of STABL. This opened up an opportunity to conduct quick turn-around computations of the transition onset point and search for the UAC parameters appropriate for further experimental and DNS studies.

Parametric stability calculations were carried out for the uncoated (solid) and coated (porous) wall. The UAC of regular microstructure comprises equally spaced vertical cylindrical blind micro-holes of fixed radius r^*, depth h^* and spacing s^*. It was shown that the coating of $r^* = 25$ μm, $s^* = 100$ μm and $h^* \approx 10r^*$ massively suppresses the second-mode instability and can lead to significant (more than twice) increase of the laminar run under Mach=7 and Mach=10 free-stream conditions. Estimates of the UAC roughness effect showed that this coating can be treated as aerodynamically smooth in the unit Reynolds number range required for transition experiments. The coating with the aforementioned r^*, s^* and $h^* = 300$ μm is recommended for testing in the CUBRC LENS I shock tunnel.

It was shown that the UAC-LFC performance strongly increases with porosity. In this connection, it is suggested to investigate a rectangular or honeycomb patterns, which allow for coatings of substantially higher porosity compared with cylindrical pores of circular cross-sections.

The porous-wall boundary-condition model does not account for end effects associated with scattering of incoming disturbances by the pore mouth and local effects on the pore bottom. Small-scale ($\sim s$) disturbances, which are generated near the UAC surface, are not considered. There is no coupling between disturbances in neighboring pores that may be not true for closely spaced cavities. These issues will be addressed in future using direct numerical simulation of the aforementioned effects.

Acknowledgements

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References

[1] Malik, M.R., Zang, T.A., and Bushnell, D.M., "Boundary Layer Transition in Hypersonic Flows," AIAA Paper No. 90-5232, 1990.

[2] Schneider, S.P., "Flight Data for Boundary Layer Transition at Hypersonic and Supersonic Speeds," *J. Spacecraft and Rockets*, Vol. 36, No. 1, pp. 8-20, 1999.

[3] Schneider, S.P., "Hypersonic Laminar-Turbulent Transition on Circular Cones and Scramjet Forebodies," *Progress in Aerospace Sciences*, Vol. 40, No. 1-2, Feb. 2004, pp. 1-50.

[4] Kimmel, R., "Aspects of Hypersonic Boundary Layer Transition Control," AIAA Paper No. 2003-0772, Reno NV, Jan. 6-9, 2003.

[5] Whitfield, J.D., and Iannuzzi, F.A., "Experiments on Roughness Effects on Boundary layer Transition up to Mach 16," AIAA Paper No. 68-0377, 1968.

[6] Batt, R.G., and Legner, H.H., "A Review of Roughness-Induced Nosetip Transition," *AIAA J.*, Vol. 21, No. 1, 1983, pp. 7-22.

[7] Lysenko, V.I., and Maslov, A.A., "Influence of Deep Cooling on Transition in Supersonic Boundary Layer," *Izv. AN SSSR, Mekh. Zhidk. I Gasa*, No. 2, 1981, pp. 43-49.

[8] Malmuth, N.D., Fedorov, A.V., Shalaev, V.I., Cole, J., Khokhlov, A.P., Hites, M., and Williams, D., "Problems in High Speed Flow Prediction Relevant to Control," AIAA Paper No. 98-2695, 1998.

[9] Fedorov, A.V., Malmuth, N.D., Rasheed, A., and Hornung, H.G., "Stabilization of Hypersonic Boundary Layers by Porous Coatings," *AIAA J.*, Vol. 39, No. 4, 2001, pp. 605-610.

[10] Rasheed, A., Hornung, H.G., Fedorov, A.V., and Malmuth, N.D., "Experiments on Passive Hypervelocity Boundary Layer Control Using an Ultrasonically Absorptive Surface," *AIAA J.*, Vol. 40, No. 3, 2002, pp. 481-489.

[11] Fedorov, A.V., Shiplyuk, A.N., Maslov, A.A., Burov, E.V., and Malmuth, N.D., "Stabilization of a Hypersonic Boundary Layer Using an Ultrasonically Absorptive Coating," *J. Fluid Mech.*, Vol. 479, 2003, pp. 99-124.

[12] Fedorov, A.V., Kozlov, V.F., Shiplyuk, A.N., Maslov, A.A., Sidorenko, A.A., Burov, E.V., Malmuth, N.D., "Stability of Hypersonic Boundary Layer on Porous Wall with Regular Microstructure," AIAA Paper No. 2003-4147, 2003.

[13] Bountin, D.A., Shiplyuk, A.N., Maslov, A.A., and Chokani, N., "Nonlinear Aspects of Hypersonic Boundary Layer Stability on a Porous Surface," AIAA Paper No. 2004-0255, Reno NV, Jan. 5-8, 2004.

[14] Maslov, A., Shiplyuk, A., Sidorenko, A., Polivanov, P., Fedorov, A., Kozlov, V., and Malmuth, N., "Hypersonic Laminar Flow Control Using a Porous Coating of Random Microstructure," AIAA Paper No. 2006-1112, Reno NV, Jan. 9-12, 2006.

[15] Egorov, I.V., Fedorov, A.V., Novikov, A.V., and Soudakov, V.G., "Direct Numerical Simulation of Supersonic Boundary-Layer Stabilization by Porous Coating," AIAA Paper No. 2007-948, Reno NV, Jan. 8-11, 2007.

[16] Kimmel, R.L., Adamczak, D., Gaitonde, D., Rougeux, A., and Hayes, J.R., "HIFiRE-1 Boundary Layer Transition Experiment Design," AIAA Paper No. 2007-534, Reno NV, Jan. 8-11, 2007.

[17] Mack, L.M., "Boundary Layer Stability Theory," Part B. Doc. 900-277, JPL, Pasadena, California, May 1969.

[18] Kozlov, V.F., Fedorov, A.V., and Malmuth N.D., "Acoustic Properties of Rarefied Gases inside Pores of Simple Geometries," *J. Acoust. Soc. Am.*, Vol. 117, No. 6, 2005, pp. 3402-3412.

[19] Jaffe, N.A., Okamura, T.T., and Smith, A.M.O., "Determination of Spatial Amplification Factors and Their Application to Predicting Transition," *AIAA J.*, Vol. 8, No. 2, 1970, pp. 301-308.

[20] Chen, F.-J., Malik, M.R., and Beckwith, I.E., "Boundary-Layer Transition on a Cone and Flat Plate at Mach 3.5," *AIAA J.*, Vol. 27, No. 6, 1989, pp. 687-693.

[21] Malik, M.R., "Hypersonic Flight Transition Data Analysis Using Parabolized Stability Equations with Chemistry Effects," *J. of Spacecraft and Rockets*, Vol. 40, No. 3, 2003.

[22] Johnson H.B., Alba C.R., Candler G.V., MacLean M., Wadhams T., and Holden M. "Boundary Layer Stability Analysis to Support the HIFiRE Transition Experiment," AIAA Paper No. 2007-311, Reno NV, Jan. 2007.

[23] Malik, M.R., Spall, R.E., and Chang, C.-L., "Effect of Nose Bluntness on Boundary Layer Stability and Transition," AIAA Paper No. 90-0112, Jan. 1990.

[24] Stilla, J., "Engineering Transition Prediction for a Hypersonic Axisymmetric Boundary layer," *AIAA J.*, Vol. 31, No. 6, 1994, pp. 1358-1364.